GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Plastics Engineering

Subject Code : DI05023051

Subject Name : Process Instrumentation and Plant Maintenance

w. e. f. Academic Year:	2026-27
Semester:	5th
Category of the Course:	PEC-04

Prerequisite:	None
Rationale:	Plastic engineering machineries and processes are now very much automated. And hence, the knowledge of instruments is essential for a plastic engineer. This subject provides the knowledge of measurement and control of plastic processing parameters. Moreover, the subject also deals with maintenance methods for machinery and helps plastic diploma holders in appreciating safety rules and does the routine maintenance. Hence this course has been designed to develop this competency and its associated cognitive, practical and affective domain learning outcomes.

Course Outcome:

The practical exercises, the underpinning knowledge and the relevant soft skills associated with this competency are to be developed in the student to display the following COs:

No	Course Outcomes	RBT Level
01	Utilize various measuring and controlling instruments.	R,U
02	Understand the fundamentals of servomechanism and PLC systems	U,A
03	Practice various types of plant maintenance such as preventive, schedule and breakdown maintenance	R,U,A
04	Manage the maintenance and safe operating procedure for different equipments.	R,A
05	Develop newer lubricating methods for prevention of wear and corrosion.	R,U,A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE(E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit-I	**Instrumentation fundamentals** 1.1 Introduction to instrumentation 1.2 Selection of measuring instruments 1.3 Errors in instruments 1.4 Block diagram of instrumentation system	05	12%
Unit-II	**Process Instrumentation** 2.1 Temperature measurement instruments and scales 2.1.1 Solid thermometer 2.1.2 Liquid thermometer 2.1.3 Gas thermometer 2.2 Thermocouple & resistance thermometer or Resistance Temperature Detector (RTD) 2.3 Pressure measurement instruments 2.3.1 Linear variable differential transformer (LVDT) 2.3.2 Pressure measurement instruments with the help of LVDT & Burdon tube 2.4 Classification of Transducers	10	22%
Unit-III	**Servo mechanisms & PLC system** 3.1 Open & close loop control system 3.2 Regulators & servo-mechanism 3.3 Servomotors 3.4 PLC control system	08	18%
Unit-IV	**Plant maintenance** 4.1 Types of maintenance 4.2 Fault finding methods 4.3 Planning & Scheduling of maintenance work 4.4 Maintenance cost and economy 4.5 Service life of equipment	06	13%
Unit-V	**Wear, Corrosion & Lubrication** 5.1 Wear types and reduction techniques 5.2 Corrosion and its types 5.3 Corrosion prevention techniques 5.4 Function and types of Lubricants 5.5 Lubrication methods	06	13%

Unit-VI	**Maintenance of Plastics processing machines** 6.1 Maintenance of an Injection Molding machine 6.2 Maintenance of Moulds 6.3 Maintenance of Hydraulic & pneumatic systems	06	13%
Unit-VII	**Safety** 7.1 Importance of safety 7.2 Major safety measures in plastics processing plants 7.3 Management responsibilities towards safety 7.4 Accidents and its prevention 7.5 Activities related to promotion of safety 7.6 Personal Protection Equipment 7.7 Different safety mechanisms in Injection and Extrusion machines	04	9%
	Total	**45**	**100%**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
36	**33**	**31**	**0**	**0**	**0**

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

Sr. No.	**Title of Book**	**Author**	**Publication with place, year and ISBN**
1	Industrial Instrumentation & Co	S.K.Singh	Publication: Tata McGraw Hill ISBN:10-0070262225 Year:2010
2	Industrial Instrumentation	Donald P.Eckman	Publication: Wiley Eastern Ltd ISBN: 9780852262061 Year:1950
3	Hand Book of Instrumentation and control	H.P.Kallen Mohren, P.	Publication: Tata McGraw Hill ISBN:10-0070332355 Year:1961

4	Plastics Industry Safety Hand Book	Dominick V.Rosato & John R. Lawrence	Publication: Van Nostrand Reinhold ISBN:10- 084361207X Year:1973
5	Industrial safety, Health and Environment Management system	R. K. Jain & Sunil S. Rao	Publication: Khanna publishers ISBN:10-8174092102 Year:2000

(b) Open source software and website:

1. www.processinst.com
2. www.automation.siemens.com/mcms/automation/en/sensor-systems/processinstrumentation/Pages/Default.aspx
3. www.scribd.com/doc/62796183/Instrumentation-Presentation
4. http://pc-education.mcmaster.ca/Instrumentation/go_inst.htm
5. http://vlabs.iitkgp.ac.in/

Suggested Course Practical List:

No.	Practical Outcomes (PrOs)	Unit No.	Approx. Hrs. required
1	Demonstrate working principle and constructional features of Bimetallic thermometer.	II	4
2	Temperature measurement using thermocouple.	II	2
3	Temperature measurement using RTD.	II	4
4	Carryout plant maintenance of Moulds.	VI	4
5	Carryout plant maintenance of Hydraulic & pneumatic systems.	VI	2
6	Prepare list of activities of preventive maintenance.	VI	2
7	Perform preventive maintenance of Injection molding machine.	VI	4
8	Find and resolve the problems in Injection molding machine.	IV	2
9	Apply lubrication to Injection Molding Machine and Extruder.	V	2
10	Carryout plant maintenance of extrusion plant.	VI	2
11.	Carryout plant maintenance of Injection molding machine.	VI	2

List of Laboratory/Learning Resources Required:

S. No.	Equipment Name with Broad Specifications
1	Thermocouple experimental setup
2	RTD experimental setup
3	Transducer experimental setup
4	Safety related precautions
5	Planning and scheduling of maintenance activity
6	Maintenance activity observation and record keeping

Suggested Project List:

1. Activities during Mold maintenance: Detailed statement indicating activities during maintenance of molds.
2. Fault finding method: Draw a decision tree and find out faulty parts.
3. Lubrication: Greasing and oiling of require parts.
4. Safety: Keep all safety precautions during plant activity.

Suggested Activities for Students:

1. Students will measurement of pressure and temperature.
2. Students will calibrate instruments.
3. Students will be greasing machines.
4. Students will remove rust from machines, molds, dies.

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